Order signed March 30, 1928. To State March 31, 1928. [36]

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**DEPARTMENT OF COMMERCE**

OFFICE OF THE SECRETARY

WASHINGTON

Answd 3/31-A

THE WHITE HOUSE MAR 30 1928 RECEIVED

March 30, 1928.

The President,
The White House,
Washington, D. C.

Dear Mr. President:

Referring to your letter dated April 7, 1927, requesting my recommendations concerning the assignment of frequencies to Government radio stations, I am enclosing for your consideration a draft of an Executive Order based upon the recommendation of the Interdepart Radio Advisory Committee, which is composed of representatives of the Departments and independent establishments interested in radio. All of the members of the Committee approved the allocation of frequencies herewith set forth, with the exception of the War Department representative, who submitted a minority report. This report was carefully considered before the enclosed Executive Order was drafted.

If this proposed Executive Order is approved by you, it is believed that the particular allocation of the frequencies to the different departments should be kept confidential until the Departments concerned decide to make them public.

The report of the full committee and the minority report of the War Department are attached hereto.

It is my understanding that this arrangement of wave lengths between the Government Departments has been approved by all of the Departments except the War Department, and the independent membership on the Commission have considered the War Department's claims are both unnecessary from a military point of view except in care of war and are greatly damaging to the commercial interests and to the interests of the other Departments.

Yours faithfully,

Herbert Hoover

HH-MM.

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# REPORT OF THE SUBCOMMITTEE ON POLICY AND LEGISLATION ON THE ALLOCATION OF FREQUENCIES FOR GOVERNMENT USE.

The Subcommittee on Policy and Legislation, in its discussions and hearings on the question of frequency assignments to Government departments, has taken for its guidance the statement of the Government Domestic Radio Policy which was adopted by the Interdepartment Radio Advisory Committee, May 5, 1925, after approval by all departments of the Government interested in radio. The principles so expressed and applicable to the present situation are paragraphs Nos. 9 and 5, which read as follows:

9. UTILIZATION OF EXISTING COMMUNICATION FACILITIES.

"No department shall erect a new station in the proximity of an existing government station, unless the same is incapable of rendering to such department the service that it requires, which shall be determined only after careful consideration by the departments concerned or by the permanent interdepartmental organization. Whenever practicable, such a situation shall be met by the expansion of the existing stations, if necessary under joint contribution of the Departments interested. No department shall close a station no longer needed by it which is serving other Government departments without first making arrangements in respect to such service that are satisfactory to the departments being served."

5. GOVERNMENT ENCOURAGEMENT TO PRIVATE RADIO ENTERPRISES.

"The Government will encourage and foster the development of privately owned and operated radio facilities in such a manner that in time of war or similar national emergency there shall be available the most effective system for the national defense. In part, this encouragement shall consist of utilizing the services of commercial stations wherever Government interests warrant in the accomplishment of the service required."

The foregoing declarations of Policy follow the general principles laid down in the President's order of 1904. A copy of the Government Domestic Radio Policy is attached hereto and marked "Exhibit A".

The Subcommittee has found that the allocations of frequencies as recommended in the report of the Subcommittee on Technical Problems referred to us on February 17th, would result in the duplication of existing Government communication facilities. As this is contrary to the radio policy quoted above, this Subcommittee considers it necessary to recommend the disapproval of requests for high frequencies for a number of proposed parallel circuits, particularly where such circuits are for overseas communication.

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This action seems particularly desirable since private enterprises desire to furnish these facilities as indicated by applications before the Federal Radio Commission from commercial companies for frequencies above 4,000 kilocycles as follows:

| Country. | Number of Stations. | Number of frequencies. |
|---|---|---|
| United States | 189 | 405 |
| Hawaii | 6 | 18 |
| Porto Rico | 2 | 4 |
| American Samoa | 1 | 2 |
| Alaska | 1 | 1 |

The Director of Posts, Philippine Islands, has stated that three frequencies above 4,000 kilocycles have been applied for by the Radio Corporation of the Philippines for communication with Hawaii and the United States.

On February 20th, the Subcommittee addressed a questionnaire to the Government agencies that had requested frequencies. A copy of this questionnaire is attached hereto and marked EXHIBIT "B".

On February 23rd, the Subcommittee heard the Coast Guard of the Treasury Department and the Bureau of Lighthouses of the Department of Commerce, on the 24th, the Communications Division of the Office of Naval Operations and on the 25th, the Signal Corps of the War Department. Each group of representatives was given an opportunity to state separately its reasons for desiring the frequencies requested and to give answers to the questionnaire referred to above. At the conclusion of these hearings each agency was requested to reconsider its application with the view to coordinating with other departments in the use of certain frequencies.

At the first hearing, the Coast Guard stated that it would withdraw its application for all frequencies above 4,000 kilocycles, namely, 4250, 8560 and 12840 kilocycles, having made arrangements with the Navy Department to communicate with the International Ice Patrol on frequencies already in use by the Navy Department for mobile service. As the Ice Patrol operations are already under way for this season, it will not be possible to change these frequencies until the end of the present Ice Patrol season, about June 30, 1928.

The Bureau of Lighthouses of the Department of Commerce has withdrawn its application for frequencies above 6,000 kilocycles except 8100 and 12150 kilocycles, which are to be used for experimental work in connection with the establishment of commercial airways in accordance with the Air Commerce Act of 1926. This withdrawal is made with the understanding that the other high frequencies originally requested be reserved for the use of the Bureau of Lighthouses should experience prove that they are necessary fo the operation of commercial airways as commercial flying increases. In the interim they are available for assignment to commercial radio services which may be willing to take them subject to six months notice that they will be required for Government use.

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The Navy Department representatives withdrew application for the following frequencies: 4005, 8010, 12015, 16020, 20025, 20075, 16300, 4355, 8710, 13065 and 17420. They expressed the hope of being able to withdraw the application for frequencies of 4525, 9050, 13575, 18100 and 22625 kilocycles if experiments now under way prove that it is possible to use 4017 kilocycles and its harmonics of 8034, 12051, 16068 and 20085 at Cavite. This frequency of 4017 kilocycles is but two kilocycles removed from the frequency (4015) now in use by the Navy at Washington. The Navy Department expressed a belief that the experiment would be successful in view of the geographical separation and difference of time.

In considering channels of overseas communication, the Subcommittee finds that at the present time the Navy Department has satisfactory and adequate circuits in operation between the United States and the Canal Zone, the United States and Hawaii, the United States and the Philippine Islands and the United States and China. According to the Navy Department's statement, these circuits are operated at not more than fifty percent capacity. The Subcommittee, therefore, does not consider it necessary for any other Government Department to parallel these circuits.

In view of the above, and especially of the willingness and ability of the Navy Department to share with other departments having overseas communication, the facilities required for the operation of the fleet, it is recommended that high frequency channels for transoceanic use by Government agencies other than the Navy Department should be disapproved. In making this recommendation, the Subcommittee, however, believes that the War Department should be assigned the necessary frequencies for communications within the Philippine Islands.

Attached hereto, marked Exhibits "C" and "D" are the War Department's application for frequencies above 4,000 kilocycles and the allocation recommended by your Subcommittee.

The Subcommittee has also recognized the advantages of grouping the frequencies allocated to Government stations. Accordingly, a statement of the allocations necessary to put this principle into effect is attached hereto and marked Exhibit "E". These assignments are for the stations designated. Slight deviations from frequencies assigned may be made provided such deviations are wholly within the CHANNEL and no interference results to other stations. EXAMPLE. "STATIONS ASSIGNED TO A FREQUENCY OF 4380 KC/S COULD UNDER THE CONDITIONS STATED BE OPERATED ON ANY FREQUENCY BETWEEN 4377.5 KC/S AND 4382.5 KC/S."

It will be noted that all applications have been approved except those involving duplications of facilities for handling overseas traffic. In making those recommendations, the Subcommittee has taken into consideration the fact that radio frequencies which are international in their effects cannot be held by the United States unless they are in actual use for a sufficient number of hours daily to prevent their appropriation by other nations.

(4)

The acquisition of rights to a frequency by any nation by the regular habitual use of such frequency is generally recognized, and the possibility of such appropriation of frequencies which the United States desires to retain can be avoided only by using such frequencies to capacity. Such use can best be obtained by making a maximum number of such frequencies available for use by private agencies in the United States, thus conserving them for the United States for use in time of national emergency.

Prior right to the use of frequencies by stations is recognized by the International Radio Convention of 1927 in Article 8 of the Convention and Article 5, Paragraph 16 of the General Regulations. These Articles are as follows:

ARTICLE 8
Interference

"The stations covered by Article 1 must, so far as practicable, be established and operated under the best conditions known to practice and must be maintained abreast of scientific and technical progress.

All stations, whatever their purpose, must so far as practicable be established and operated in such a way as not to interfere with the communications or radio services of other contracting Governments and of individuals or of private enterprises authorized by these contracting Governments to carry on public radio communication service."

ARTICLE 5, Paragraph 16.

"The frequencies assigned by Administrations to all new fixed land or radio broadcasting stations which they may have authorized or of which they may have undertaken the installations must be chosen in such a manner as to prevent as far as possible interference with international services carried on by existing stations the frequencies of which have already been notified to the International Bureau. In a case of change of frequency of an existing fixed land or broadcasting station, the new frequency assigned to this station must comply with the above conditions.

The interested Governments shall agree, in case of need upon the fixing of waves to be assigned to the stations in question as well as upon the conditions for the use of waves so assigned. If no arrangement intended to eliminate interference can be arrived at, the provisions of Article 15 of the Convention may be applied."

We recommend the approval of the allocations submitted by the Sub-Committee on Technical Problems below 4,000 kilocycles with the addition of the frequencies between 3340 and 3410 kilocycles desired by the Bureau of Lighthouses of the Department of Commerce and the

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further additional frequencies requested by the War Department for assignment to stations as follows: 166.6 kilocycles and 93.7 kilocycles for stations WXZ, Nulato, Alaska; 250 kilocycles, Sitka, Alaska; 258 kilocycles; Wrangell, Alaska; 266 kilocycles, Petersburg, Alaska; 273 kilocycles, Cordova, Alaska; 284 kilocycles, Haines, Alaska.

C. R. P. Rodgers

W. R. Vallance

A. E. Cook

C. B. Jolliffe

March 8, 1928.

Amended report as adopted March 13, 1928.
(Amendments in caps)

(6)

High & Low [136]

Total Frequencies - 411

| | |
|---|---|
| Army | 140 |
| Navy | 157 |
| Commerce | 101 |
| C. Guard. | 10 |
| Interior | 1 |
| | 409 |
| All Depts. for distress Mil. oprs. | 1 |
| Total - | 411 |

(2)

# STATEMENT OF GOVERNMENT DOMESTIC RADIO POLICY.
## ADOPTED BY THE INTERDEPARTMENT RADIO ADVISORY COMMITTEE.

Exhibit A

May 8, 1925.

1. Introduction.

The following statement of policy gives only general principles. In dealing with any specific situation, the application of these principles may be so manifest that the particular situation under consideration may be decided without the necessity for conference and compromise. In all other cases the parties at interest should confer, preferably through the interdepartmental organization hereinafter provided for, and arrive at decisions based upon these principles.

2. Interest of the Government in Radio Communication.

The Government's interest in Radio Communication is paramount for national defense. The Government is charged with the regulation of Radio Communication as a means for rendering a public service. The Government uses Radio Communication in the administration of its business.

Appendix I contains illustrations of the ways in which the several Government agencies are concerned with Radio Communication.

Appendix II contains illustrations of the ways in which private interests are concerned with radio communications.

3. Coordination of government radio facilities.

Coordination of government owned and operated facilities must take into consideration the authority of the several departments and governmental agencies interested, and such coordination is exercised by the President aided by a coordinating agency in which the interested departments and governmental agencies have representation. This coordinating agency shall consist of representatives of all the departments of the government and of such other governmental agencies as have sufficient interest in radio to warrant their participation. The functions of this committee shall be as provided herein.

4. Regulation of private radio facilities.

In order that the greatest good may be accomplished with the minimum of interference and conflict in the utilization of radio facilities, it is necessary that such utilization be regulated and coordinated. Regulation of privately owned and operated radio facilities is by law laid upon the Department of Commerce and the Interstate Commerce Commission. In order that such regulation may be coordinated with Government utilization, the Department of Commerce should exercise its regulative function in conformity with the principles herein set forth.

5. Government encouragement to private radio enterprises.

The Government will encourage and foster the development of privately owned and operated radio facilities in such a manner that in time of war or similar national emergency there shall be available the most effective system for the national defense. In part, this encouragement shall consist of utilizing the services of commercial stations wherever government interests warrant in the accomplishment of the service required. Government stations shall not solicit nor, except in an emergency, shall they accept non-government radio traffic from non-government stations where commercial radio stations can handle it satisfactorily. The Government, on the other hand, will facilitate the use of its service by non-government interests whenever such employment does not interfere with the government business, when non-government radio systems are not capable of meeting the normal communication requirements.

6. Prorogative of government departments as to radio policy.

Subject to the limitations stated herein and to such instructions as may be issued by the President each of the Executive Departments shall be the sole and final judge as to its own policy in respect to radio matters; shall determine its own mission; and decide upon the steps toward executing such mission. Such decisions shall include the questions of location of stations, power output, apparatus to be installed, service to be rendered, the capital investment and operating details.

7. Coordination between government departments.

Before deciding upon any changes in location or services rendered, each department shall ascertain through the existing interdepartmental coordinating organization how such proposed changes will affect the interests of any other department, and shall in its decisions, give the suggestions from other departments careful consideration, in order, that the final results will produce a government radio system designed to meet the needs of the Federal Government as a whole in war as well as in peace.

8. Extension of services to other government departments.

The use of government radio facilities shall be extended by the department operating such facilities to other departments of the government insofar as can be done without impairing the efficiency of the service of the operating department, subject to its regulations. If the additional traffic can be handled with the funds and personnel available to the operating department, no charge will be made for such service. If the traffic offered requires personnel and funds beyond the resources of the operating department the expenses will be shared by the beneficiaries on a basis mutually agreeable to the departments concerned.

Radio Policy.

9. Utilization of existing communication facilities.

No department shall erect a new station in the proximity of an existing government station, unless the same is incapable of rendering to such department the service that it requires, which shall be determined only after careful consideration by the departments concerned or by the permanent interdepartmental organization. Whenever practicable, such a situation shall be met by the expansion of the existing stations, if necessary under joint contribution of the Departments interested. No department shall close a station no longer needed by it which is serving other government departments without first making arrangements in respect to such service that are satisfactory to the departments being served. If no agreement can be reached through the interdepartmental coordinating organization the matter must be appealed to the Chief Coordinator and, if necessary, through the Director of the Budget, to the President.

10. Joint contribution agreement.

Where stations are operated under joint contribution agreement, such agreements shall provide that one of the departments concerned shall exercise the responsible control. It is considered undesirable to mix civilian and enlisted operating personnel, particularly at stations so situated that resident quarters have to be provided on the station reservation for such personnel.

11. Function of the interdepartmental organization.

Measures to prevent interference; the assignment of frequencies within the allotted wave bands; determination of the character of broadcast material; and other radio matters of common interest to all departments, not having an international aspect, shall be determined through the permanent interdepartmental organization. Departments which operate radio stations shall give priority to calls of distress and to other messages concerning safety of human life or maring or aerial navigation, but shall otherwise determine priority of service and other features involving the details of traffic.

12. Coordinating Supervision of government radio facilities.

Cooperation and coordinated supervision as prescribed under paragraph 3 above, is, under existing circumstances, the only practicable means of realizing the highest degree of efficiency in the utilization of government radio facilities. The coordination at present exercised through the Interdepartment Radio Advisory Committee is advisory. It provides an effective means of investigation and presentation of pertinent facts for consideration by the President of the merits of any case which requires his action.

Radio Policy.

13. Centralized supervision of government radio facilities.

Centralized control of Government Radio communications, by means of consolidation or otherwise, is not at present either practicable or desirable. The requirements of the several operating departments differ so radically that no one department can completely fill the needs of another without an unreasonable increase of Federal expenditures. Those departments which are now users but not operators of government owned facilities can have their requirements supplied in large measure by the present operating departments, but conditions may arise where it may be necessary to provide installations where none now exist and which are not required by the present operating departments.

14. Settlement of Questions between Government Agencies Relative to Radio Communication.

Where only two agencies are concerned in any question arising, such question or questions can best be settled through direct conference. In all other cases settlement should be reached through the instrumentality of the interdepartmental organization.

15. Radio Communications as an Element of National Defense.

Insofar as radio communication is an element of National Defense, the same shall be under the control of the War and Navy Departments jointly exercised through such agency as may be established by these departments. The recommendations of the Joint Board Serial No. 183, dated the 16th of January, 1923, are approved. For information, these recommendations are given in Appendix III.

16. Radio Communication on the Panama Canal.

Radio communication on the Panama Canal Zone is considered primarily a question of National Defense and is to be treated generally as prescribed in the preceding paragraph, and in Appendix III, or in such manner as is approved by the War and Navy Departments.

17. Radio in Undeveloped Territories.

The government is recognized as having a certain responsibility to provide facilities for undeveloped territories, such as Alaska, both for effective administration of the territory and as an aid to its economic development. Except for radio stations necessary for the National Defense, the government may properly choose to provide the facilities by equitable agreement with commercial operating companies in preference to direct governmental operation.